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Permo–Triassic palaeomagnetism in the Emei Mountain Region, southwest China

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SUMMARY

A palaeomagnetic study has been made of samples collected from Permo–Triassic type sections in the Emei Mountain region of southwest China. No results were obtained from the Lower Permian Qixia Formation limestones or the (Lower)–Upper Permian Maokou Formation limestones. The Upper Permian (Longtanian) Emeishan Basalt Formation shows excellent stability to both alternating field and thermal demagnetization. Results from 24 flows (125 samples) are all of normal polarity, have a positive fold test (Cretaceous or Tertiary folding), and give a pole position at 54.5°N, 254.3°E with $dp = 1.8°$, $dm = 3.5°$. The basalts were thus presumably extruded after the end of the Kiaman Reversed Superchron. The overlying Upper Permian Xuanwei Formation of red mudstones and sandstones (Changxingian in age) also shows a positive fold test from $N = 12$ samples after thermal treatment and has a positive reversals test at level *C*. The pole position is close to that of the Emeishan Basalt and lies at 57.0°N, 260.9°E with $dp = 4.3°$, $dm = 8.5°$. The Lower Triassic (Griesbachian to mid-Nammalian) Feixianguan Formation is primarily a red sandstone sequence at this locality. Thermal treatment of the collection of 114 samples reveals two main components. The *A* component is always observed below 600 °C and probably represents a DRM either from magnetite or specular haematite. The *B* component resides in haematite and is often of the opposite polarity superimposed on the *A* component. It is revealed only above 600 °C and is a CRM acquired during diagenesis. Although normal components dominate both, the *A* and *B* components also have reverse polarities. The normal *A* and normal *B* components are statistically indistinguishable and the normal *B* component shows a positive fold test. The reverse *B* component is, however, not quite antiparallel to the normal *B* component and may therefore be a second CRM acquired at a slightly different time to the normal *B* component. All these components are thought to have arisen during deposition and diagenesis over a few million years as has been observed in other studies of redbeds. The dominance of normal components is consistent with the observation that the earlier part of the Griesbachian Stage of the Lower Triassic is mainly a normal polarity chron. The overall mean of the $N = 101$ both normal and reverse *A* and *B* components is considered to be the best estimate of the palaeofield in the Early Triassic giving a pole position at 56.2°N, 227.6°E with $dp = 1.7°$, $dm = 3.4°$. This pole lies 15.2° away from the Emeishan Basalt pole suggesting a polar shift occurred near the Permo–Triassic boundary. A review of Late Permian and Early Triassic poles from the South China Block shows a similar polar shift.

Key words: China, magnetostratigraphy, palaeomagnetism, Permian, tectonics, Triassic.

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INTRODUCTION

It is now generally recognized that Asia and especially China is made up of a number of blocks that have accreted onto a Eurasian core with subsequent deformation as the successive accretion of further blocks has taken place (Hamilton 1970; Burrett 1974; Zhang, Liou & Coleman 1984; Tapponnier, Peltzer & Armijo 1986). The first palaeomagnetic results from China using modern techniques by McElhinny *et al.* (1981) showed that in Permian times both the North China Block and the South China Block were widely separated from Eurasia and from each other. Numerous studies have subsequently enhanced and confirmed those initial results. In the original McElhinny *et al.* (1981) paper, we reported results from the Emeishan Basalts in the Emei Mountain region south of Chengdu on the edge of the Sichuan Basin. The results from the basalts were a part of an overall study of the type sections for the Permo–Triassic in South China. In this paper we report the final results from our study of the entire sequence in this region.

Figure 1 shows a map of the various blocks of Asia. The original program was a joint project between the palaeomagnetic laboratories of the Chinese Academy of Geological Sciences in Beijing, the Research School of Earth Sciences at the Australian National University (ANU) in Canberra and at the CSIRO Division of Exploration Geoscience in Sydney to study the Permo–Triassic type sections of the North China Block and the South China block. Initial results from these studies were published in McElhinny *et al.* (1981) and complete results from the North China Block sections will appear later (Embleton *et al.*, in preparation). A brief report of the overall results from North and South China has previously been given in Chinese by Ma & Zhang (1989). A study of Permo–Triassic sections in the northern part of the Tarim Block that formed part of this China–Australia cooperative program has been published by McFadden *et al.* (1988).

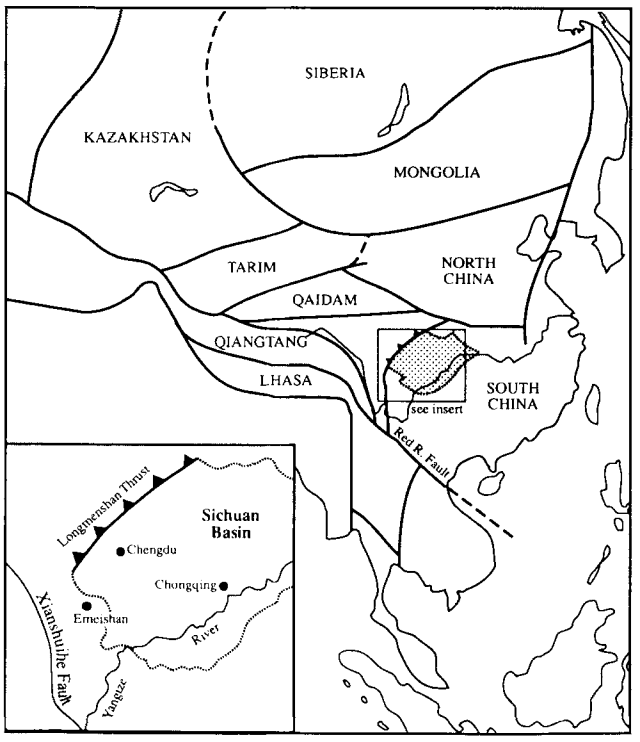


Figure 1. Simplified map showing the major blocks of Asia. The sampling area at Emeishan is shown in the inset.

GEOLOGY AND SAMPLING

The type section for the Permo–Triassic in South China is located at Lianghekou (29.6°N, 103.4°E) near Emei Mountain (Emeishan) south of Chengdu and just off the edge of the Sichuan Basin, but well away from the Longmenshan Thrust that forms the northeast boundary of the basin (see inset to Fig. 1). A further section at Liujinggou (29.4°N, 103.5°E) was also sampled in order to obtain a fold test on the various formations. Table 1 gives a general outline of the stratigraphy in this region. Palaeomagnetic samples were collected with a portable sampling drill and oriented with a sun compass through both sections of Liujinggou and Lianghekou.

The basic Permian (Asselian) Liangshan Formation of continential sandstones is only about 1 m in thickness and is overlain by limestones of the Qixia Formation (100 m thick) of Sakmarian to Artinskian age. The overlying Maokou Formation (320 m thick) ranges from Kungurian through to the Capitanian (see Harland *et al.* 1990). These limestones contain abundant Permian fusilinids, corals and brachiopods. At Lianghekou 13 samples were collected from the Maokou formation limestones where the beds dip at 75° towards 70° (NE). A further 37 samples were collected at Liujinggou where the beds dip between 35–43° towards 140° (SE). In addition 19 samples were collected from the Qixia Formation at Liujinggou with dips from 40–45° towards 142° (SE).

A disconformity separates the Maokou Formation from the Emeishan Basalt Formation of Late Permian age. The basalts have well developed amygdaloidal texture and columnar joints. The main minerals are plagioclase and augite with no olivine. No pillow structure is seen and the lavas are considered to be continental basalts. K-Ar ages on the basalts range from 219 to 253 Ma (Yuan & Li 1985) with a tendency towards older ages to the south. A mean age of 235 Ma is cited for flows from Sichuan Province. The basalts underlie the Xuanwei formation (Changxingian) and overlie the Maokou Formation (Kungurian to Capitanian). Thus on

Table 1. Generalized stratigraphic sequence of the Permo–Triassic in the Emei Mountain region, southwest China.

Lower Triassic (Griesbachian to Nammalian)	Feixianguan Formation (Red sandstone sequence)
Upper Permian (Changxingian)	Xuanwei Formation (Red claystones, sandstones with thin coal seams)
Upper Permian (Longtanian)	Emeishan Basalt Formation (Amygdaloidal basalts)
(Lower)-Upper Permian (Kungurian-Capitanian)	Maokou Formation (Limestones)
Lower Permian (Sakmarian-Artinskian)	Qixia Formation (Limestones)
Lower Permian (Asselian)	Liangshan Formation (Sandstones)

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purely stratigraphic considerations the basalts are confined to be within the Longtanian Stage on the 1989 Geologic Time Scale of Harland *et al.* (1990). 17 flows (85 samples) were collected in the Lianghekou section through a thickness of 250 m. Here the lowest 11 flows dip at 77° towards 64° whereas the higher six flows dip at 87° towards 62° (NE). A further seven flows (36 samples) were collected in the Liujinggou section where the flows dip at 46° towards 125° (SE).

The Xuanwei Formation disconformably overlies the Emeishan Basalt Formation and at the Lianghekou section consists of purplish-red claystone interbedded with purplish-grey to dark-red medium-grained sandstone with thin coal seams. Fossil floras indicate the Xuanwei Formation is latest Permian in age (Changxingian). 17 oriented samples were collected from a thickness of 40 m starting immediately above the basalts. Four samples were collected from a red mudstone band at the base of the formation just above the basalts. Here the beds dip at 87° towards 62° (NE). A further 13 samples were collected 25 m higher in the sequence from a 15 m section of red claystones and sandstones that are slightly overturned dipping at 97° towards 57° (NE).

Conformably overlying the Xuanwei Formation is the Feixianguan Formation, a red sandstone sequence that is Early Triassic in age ranging from Griesbachian to mid-Nammalian (Harland *et al.* 1990). The boundary between the Xuanwei Formation and the Feixianguan Formation is taken to be the Permo–Triassic boundary. 114 oriented samples were collected through a thickness of 135 m in the Feixianguan Formation at Lianghekou probably representing the lower half of the formation. Here the beds are again slightly overturned with dips varying from 94° to 101° through the sequence towards 65° (NE). The age of folding is either Cretaceous or Tertiary and deformation in this region is probably associated with the collision of India with Asia (Tapponnier *et al.* 1986). A full discussion of tectonic activity in the region is given in Enkin *et al.* (1992).

PALAEOMAGNETIC MEASUREMENTS

Laboratory measurements were carried out at the palaeomagnetic laboratory at the ANU in Canberra. Magnetic measurements were made using a Digico spinner magnetometer. All samples were subjected to stepwise thermal demagnetization up to 670 °C in the case of the red sediments, and up to 570 °C in the case of the basalts and limestones using the apparatus described by McElhinny, Luck & Edwards (1971). Some basalt samples were also treated in alternating magnetic fields using the method described by McElhinny (1964); however, thermal demagnetization seemed to be the most efficient cleaning process. Data processing was carried out at the palaeomagnetic laboratory of the CSIRO in Sydney using orthogonal vector diagrams following Zijderveld (1967) and these were analysed using the principal-component analysis method of Kirschvink (1980) and the linearity spectrum analysis method of Schmidt (1982).

Feixianguan Formation

Thermal demagnetization was carried out in 12 steps up to 670 °C. The samples often showed complex behaviour above 600 °C and examples of the various types of behaviour are illustrated by the Zijderveld diagrams in Fig. 2. The simplest behaviour shows that apart from various temporary (laboratory or transport induced) components eliminated at low temperatures there exists a single component of either normal (Fig. 2a) or reverse polarity (Fig. 2b) that decays directly to the origin at the Curie Temperature of haematite. We shall refer to this high-temperature component observed above 600 °C as the *B* component. In some cases, illustrated especially in Fig. 2(d), a component of almost opposite polarity (reverse in this case) but with different inclination is removed between 200° and 400 °C. Above 450 °C the characteristic component (normal polarity) decays simply to the origin. Fig. 2(c) shows a case with a relatively small reverse component being removed at low temperatures, whereas in Fig. 2(d) the reverse component being removed is relatively large. More complicated situations are illustrated in Figs 2(e) and 2(f). In these two cases the first component to be removed is of normal polarity and this persists up to 550 °C. We shall refer to this component below 600 °C as the *A* component. In the cases illustrated there is in addition a *B* component of opposite polarity that is only observed above 600 °C. Fig. 2(e) shows a rare case in which the reverse polarity *B* component above 600 °C is much larger that the normal polairy *A* component removed below 600 °C. Fig. 2(f) is the more typical situation in which a small reverse *B* component is revealed after removal of a large normal polarity *A* component below 600 °C. Although on a few occasions a reverse polarity *A* component was observed, there were no situations in which a reverse polarity *A* component was removed at temperatures up to 600 °C to reveal an underlying normal polarity *B* component above 600 °C. However, because the *A* component is usually very much larger in magnitude compared with the *B* component, the *B* component could often not be determined with any accuracy after removal of a large *A* component. In many cases the *A* component being removed up to 600 °C was clearly not decaying to the origin, but the intensity had dropped so low above 600 °C that the high-temperature *B* component (presumably of opposite polarity) could not be determined.

Xuanwei Formation

Samples from this formation showed much simpler characteristics during thermal demagnetization compared with the Feixianguan Formation. Demagnetization was carried out in eight steps commencing at 300 °C up to 675 °C. Both normal and reverse components were observed but in all cases the samples showed a single component decaying to the origin of the Zijderveld diagrams as illustrated in Fig. 3.

Emeishan Basalt

Both alternating field and thermal demagnetization were carried out on samples from the Emeishan Basalt Formation using separate specimens from the same sample. In some cases thermal treatment was used following alternating field demagnetization. Alternating field demagnetization was carried out in 10 steps up to 55 mT and thermal demagnetization in nine steps up to 550 °C. Both methods

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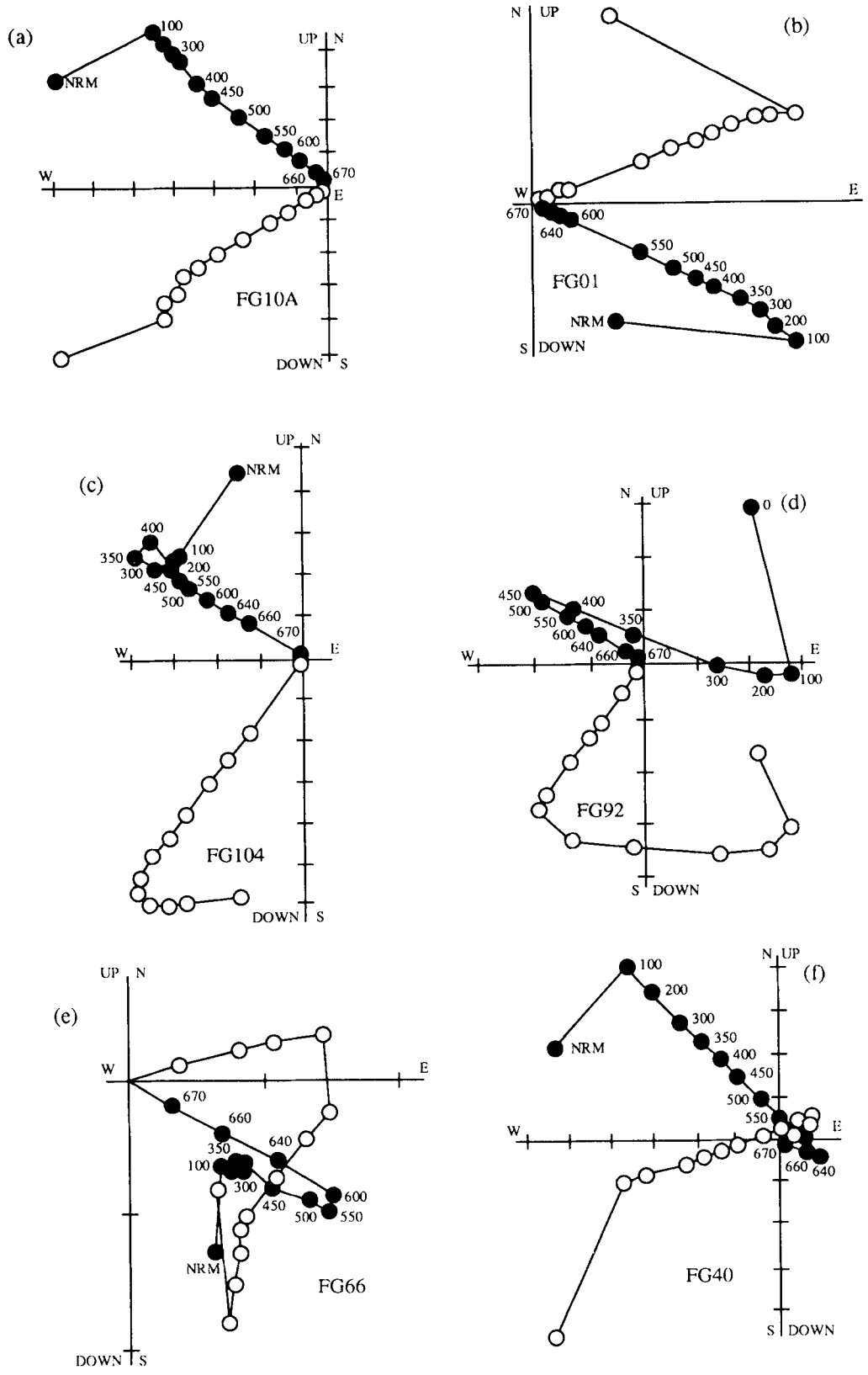


Figure 2. Examples of Zijderveld (1967) vector plots for samples from the Feixianguan Formation during thermal demagnetization. Vectors are plotted in geographic coordinates with solid circles in the horizontal plane and open circles in the vertical plane. (a) Single normal high-temperature component *B*. (b) Single reverse high-temperature component *B*. (c) Normal high-temperature component *B* with small reverse component *A* removed between 200 and 400 °C. (d) Normal high-temperature component *B* with large reverse component *A* removed between 200 and 450 °C. (e) Reverse high-temperature component *B* with normal component *A* removed between 350 and 550 °C. (f) Reverse high-temperature component *B* with large normal component *A* removed between 100 and 600 °C.

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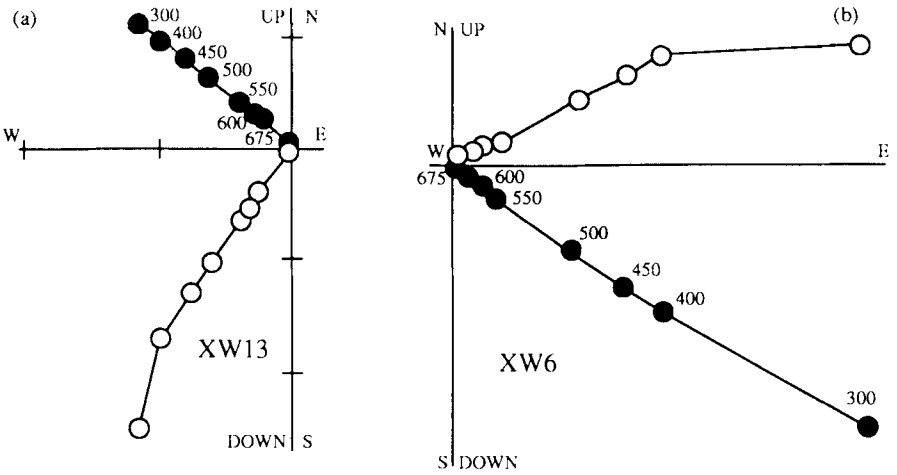


Figure 3. Typical Zijderveld (1967) vector plots for samples from the Xuanwei Formation during thermal demagnetization. Geographic coordinates used with symbols as in Fig. 2. (a) Normal magnetization. (b) Reverse magnetization.

showed that apart from temporary components removed in low alternating fields or temperatures, the magnetization consisted of a single vector that decayed to the origin of the Zijderveld diagrams. Examples are illustrated in Fig. 4.

Fig. 4(a) shows the result of alternating field treatment on a sample from a flow from the Lianghekou Section and 4(b) shows the result from a sample from an adjacent flow in which alternating field treatment to 55 mT is followed by thermal demagnetization in 500° and 550 °C. Figs. 4(c) and (d) show results of alternating field and thermal demagnetization on samples from the same flow in the Liujinggou Section. Thermal demagnetization was the preferred treatment for most samples because less scatter was encountered although the differences were very minor.

The variation of magnetic susceptibility from liquid nitrogen temperature up to room temperature was measured for seventeen representative samples of the basalts and then classified into the three types of behaviour noted by Senanayake & McElhinny (1981). Nine of the samples were of Type 2 in which the susceptibility gradually increases with decrease in temperature. In these samples this would be interpreted to indicate that the main magnetic mineral is a titanomagnetite containing exsolved ilmenite lamellae such that the subdivided regions, mainly of magnetite composition, act as independent single-domain grains. This observation is consistent with the excellent stability to alternating field and thermal demagnetization observed in so many samples. The remaining eight samples were all of Type 3 showing a peak in susceptibility with decreasing temperature with the value finally decreasing to below its room temperature value at liquid nitrogen temperature. On the classification of Senanayake & McElhinny (1981) these samples contain mostly multidomain or pseudo-single-domain magnetite.

Maokou and Qixia Limestones

Samples from both these formations were generally weakly magnetized. Unfortunately all samples were initially treated in a temperature of 300 °C in error virtually wiping out any possibility of proper analysis of Zijderveld diagrams. At higher temperatures the magnetization became too weak to measure with any accuracy and thus we have been unable to make any conclusions about the palaeomagnetism of these formations.

DISCUSSION OF PALAEOMAGNETIC RESULTS

A summary of the overall paleomagnetic results from the Feixianguan, Xuanwei and Emeishan Basalt Formations is given in Table 2.

Emeishan Basalt

In Table 2 the 24 flows have been assigned to three groups corresponding to the three different dips observed in the basalts labelled as Limbs 1, 2 and 3. In the Lianghekou Section the *in situ* directions for Limbs 1 and 2 are close to the axial-dipole field direction (0°, +48.6°) and the present field direction (359°, +43°) as shown in Fig. 5(a). Nevertheless the *in situ* means from both limbs are significantly different from both the axial-dipole field and present-field directions. For Limb 1 the differences are 11.4° and 8.5° respectively which are both greater than the 95 per cent confidence limit of 5.5°. For Limb 2 the differences are 13.4° and 11.1° respectively, again both greater than the 95 per cent confidence limit of 7.5°. The Limb 3 mean clearly differs markedly from both the axial dipole and present fields (Fig. 5a) and on restoring the three limbs to the horizontal by simple rotation about their strike the directions from all three limbs converge to an overall mean direction with declination 16.4° and inclination −4.5° (Fig. 5b and Table 2). This is clearly a positive fold test.

The significance of the fold test on the Emeishan Basalt can be tested in a variety of ways. Using the McElhinny (1964) fold test the ratio of precisions for all 24 flows gives $k2/k1 = 11.5$ and this is significant at the 99 per cent confidence level. Using the McFadden & Jones (1981) test for $N = 24$ flows and $m = 3$ limbs the statistic P, where

$$P = \frac{(N-m)}{(m-1)} \times \frac{\Sigma R_i - (R^2/\Sigma R_i)}{2(N-\Sigma R_i)}$$

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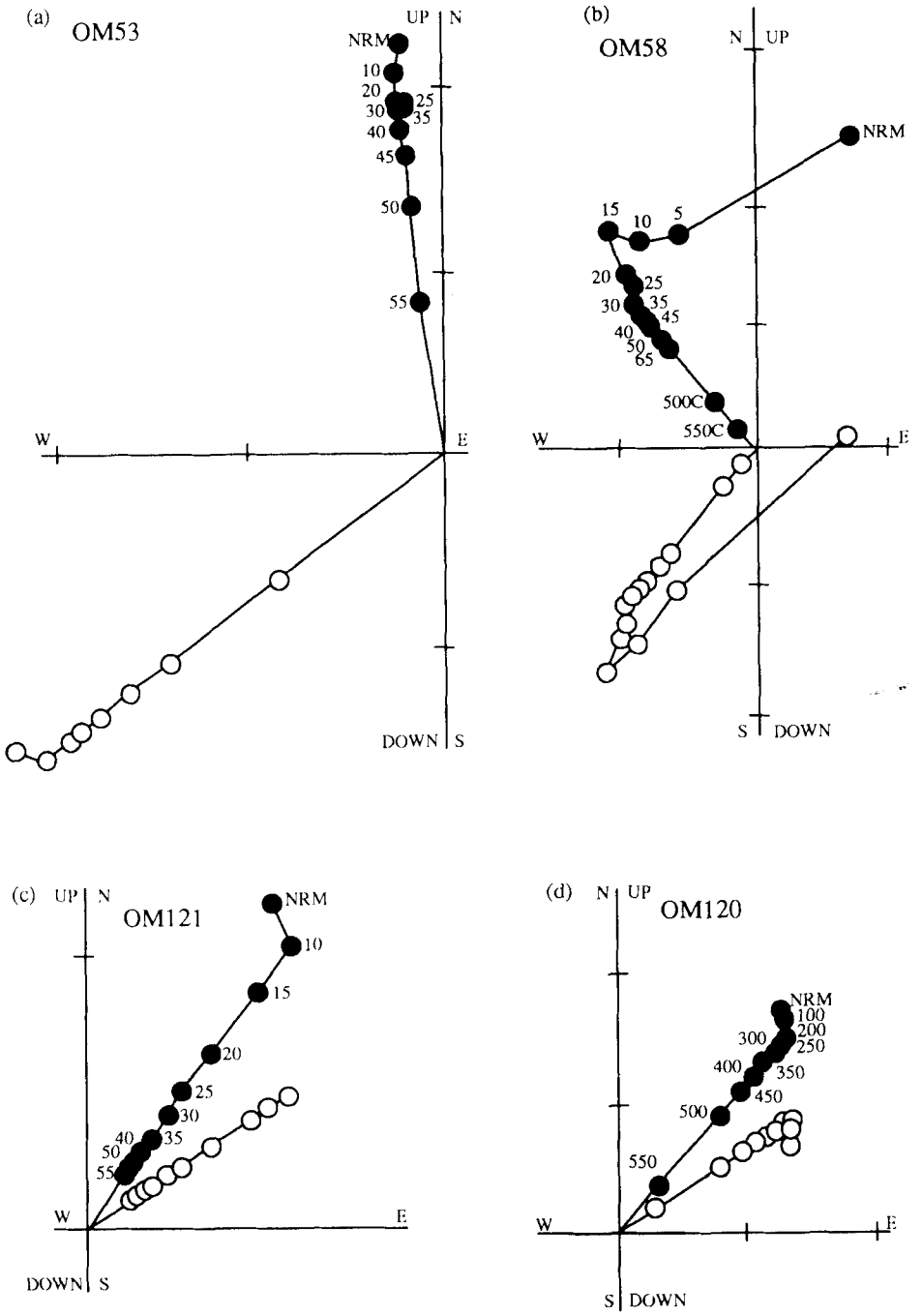


Figure 4. Typical Zijderveld (1967) vector plots for samples from the Emeishan Basalts during alternating field (a–c) and thermal (d) demagnetization. Alternating fields given in millitesla. (b) shows the results of thermal demagnetization at 500 and 500 °C after initial alternating field demagnetization up to 65 mT. Geographic coordinates used with symbols as in Fig. 2.

is compared before and after unfolding using F-ratio tables with $2(m-1)$ and $2(N-m)$ degrees of freedom. R is the length of the overall resultant vector and R_i are the lengths of the resultant vectors from each of the m limbs. Before unfolding the statistic, P has a value 114.9, which is well above the 95 per cent significance point of 2.60 and the 99 per cent significance point of 3.83 for $F_{4,42}$. The hypothesis of a common mean direction *in situ* may be thus rejected. After unfolding p becomes 1.12 so that there is no reason to suppose that the limbs do not have a common mean direction. Again the fold test is significant at the 99 per cent confidence level. Alternatively the new fold test of McFadden (1990) can be used. McFadden's statistic ξ has critical 95 per cent value of 5.702 and 99 per cent value of 8.007 for $N = 24$. With the directions *in situ* $\xi = 22.936$ so the directions are clearly correlated with tectonic dip, whereas after unfolding $\xi = 1.390$ and the directions have no apparent correlation with dip. Once again the fold test is significant at the 99 per cent confidence level.

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Table 2. Directions of magnetization observed in Permo-Triassic rocks in the Emei Mountain region (29.6N, 103.4E).

Component	N*	D_g	I_g	D_s	I_s	R	k	α_{95}
Feixianguan								
A Component *N*	66	308.9	47.0			62.814	20.4	4.0
	66			31.2	14.8	62.535	18.8	4.1
R	2	128.9	-36.1			1.953	21.4	
	2			199.6	-22.7	1.944	17.8	
Mean	68	308.9	46.7			64.733	20.5	3.9
	68			30.9	15.0	64.427	18.8	4.1
B Component *N*	22	310.9	42.8			21.165	25.1	6.3
	22			25.6	15.1	21.365	33.1	5.5
R	11	131.4	-28.6			10.412	17.0	11.4
	11			190.9	-17.8	10.312	14.5	12.4
Mean	33	311.1	38.1			31.364	19.6	5.8
	33			20.9	16.1	31.458	20.8	5.6
Combined AB								
Normal	88			29.8	14.9	83.829	20.9	3.4
Reverse	13			192.3	-18.6	12.234	15.7	10.8
Overall Mean	101	309.6	43.9			95.849	19.4	3.3
	101			27.7	15.6	95.665	18.7	3.3
Xuanwei								
Normal	8			17.4	- 2.4	7.785	32.6	9.9
Reverse	4			181.3	2.9	3.908	32.5	16.4
Limb 1	4	347.2	40.9	14.3	- 9.4	3.987	231.8	6.0
Limb 2	8	318.9	43.3	10.9	1.0	7.651	20.0	12.7
Overall Mean	12	328.8	43.3			11.461	20.4	9.8
	12			12.0	- 2.6	11.590	26.8	8.5
Emeishan Basalt								
Limb 1	11	347.9	41.1	17.0	- 1.6	10.855	68.8	5.5
Limb 2	6	344.2	41.2	14.1	- 7.2	5.938	81.3	7.5
Limb 3	7	27.5	-17.3	17.4	- 6.7	6.915	70.4	7.3
Overall Mean	24	0.4	26.3			20.261	6.2	13.0
	24			16.4	- 4.5	23.677	71.2	3.5

* Samples for the Feixianguan and Xuanwei Formations, flows for the Emeishan Basalt Formation.

D_g, I_g and D_s, I_s are declination, inclination in geographic and stratigraphic coordinates respectively. R is the resultant of N unit vectors, k is the Fisher (1953) precision and α_{95} the radius of the circle of 95 per cent confidence about the mean direction.

Xuanwei Formation

The Xuanwei Formation has both normal and reverse directions of magnetization. The results, summarized in Table 2 and illustrated in Fig. 6(a), show that the normal and reverse means are 163.9° apart. Using the Reversals Test of McFadden & McElhinny (1990), the data indicate a positive test at Level *C*. For the overall mean there is a small increase in precision from $k1 = 20.4$ to $k2 = 26.8$. This increase is not large enough to pass the McElhinny (1964) fold test. However, the samples can be subdivided into two limbs that differ in dip by 10°. The four mudstone samples at the base of the formation represent one limb with mean direction $D = 347.2$, $I = 40.9$ *in situ* and $D = 14.3$, $I = -9.4$ after unfolding with $k = 231.8$ and $\alpha_{95} = 6.0°$ ($N = 4$). The eight sandstone samples that are stratigraphically higher represent another limb with mean direction $D = 318.9$, $I = 43.3$ *in situ* and $D = 10.9$, $I = 1.0$ after unfolding with $k = 20.0$ and $\alpha_{95} = 12.7°$ ($N = 8$). Unfortunately the fold test of McFadden & Jones (1981) cannot be applied because the precisions for the groupings from the two limbs differ significantly. However, Fig. 6(b) shows that two means differ by 21.0° *in situ* and their circles of 95 per cent confidence do not overlap. After unfolding the means now differ by only 11.0° and the circles of confidence now overlap (Fig. 6b). Using the two 95 per cent circles of confidence an approximate significance point of $(12.7^2 + 6.0^2)^{1/2}$ or 14.0° can be calculated for the two mean directions. This suggests that the two limbs have significantly different mean directions *in situ* but are statistically indistinguishable after unfolding. This is precisely the form of test that McFadden & Jones (1981) use for the case of two limbs having the same precision. Thus a positive fold test is clearly indicated. The new McFadden (1990) fold test confirms the above approximate calculation. The statistic ξ has critical value of

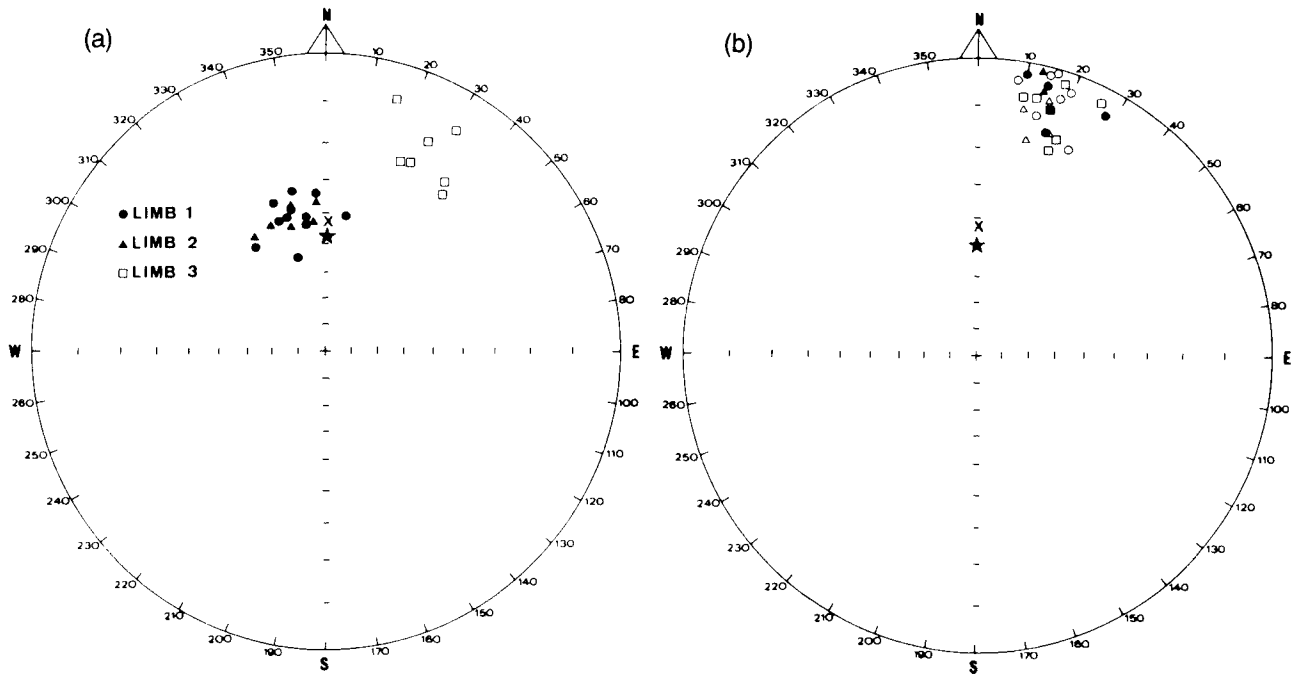


Figure 5. Equal-angle stereographic projection of site mean directions of magnetization from the Emeishan Basalt. The star is the present axial dipole and the cross the present Earth's field. Solid (open) symbols are plotted on the lower (upper) hemisphere. (a) Geographic coordinates. (b) Stratigraphic coordinates.

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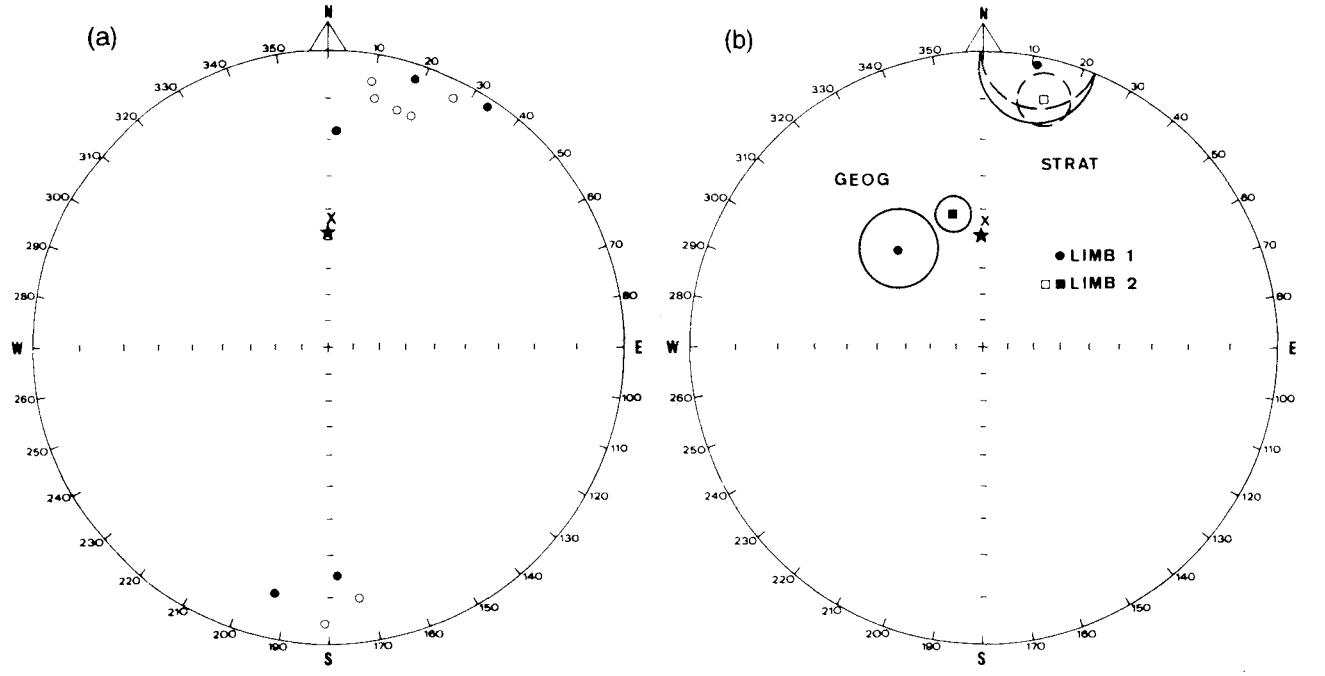


Figure 6. Equal-angle stereographic projection of directions of magnetization from the Xuanwei Formation. (a) Sample mean directions in stratigraphic coordinates. (b) Mean directions from the two bedding limbs with their 95 per cent circles of confidence in geographic and stratigraphic coordinates. Symbols as in Fig. 5.

4.036 at the 95 per cent confidence level and 5.624 at the 99 per cent level for $N = 12$. The directions *in situ* give $\xi = 5.859$ so that there is a clear correlation with tectonic dip, whereas after unfolding $\xi = 3.794$ and there is no apparent correlation with dip. Although the dip of the two limbs differs by only 10°, fortuitiously the direction of unfolding is such as to maximize the convergence of the two separated means after unfolding.

Feixianguan Formation

Analysis of Zijderveld diagrams showed both the *A* and *B* components have normal and reverse vectors. The statistics for these components are listed in Table 2. The *B* component clearly resides in haematite whereas the *A* component could reside in magnetite or haematite. When they occur in the same sample they are often of opposite polarity and are not strictly antiparallel to each other. This strongly suggests that they were acquired at different times. However, the various analyses given in Table 2 show the following interesting characteristics. The means of the normal *A* and *B* components after structural correction differ by only 5.4° and are thus statistically indistinguishable. The reverse *B* component lies 165.7° from the normal *B* component and the statistical parameters show that it fails the reversals test of McFadden & McElhinny (1990). As only two samples gave a reverse *A* component a reversals test for this component is inconclusive. Most interesting, however, is that there is an increase in the Fisher (1953) precision parameter for the 22 normal *B* component measurements on structural correction. The ratio of the precisions gives $k2/k1 = 1.319$ corresponding to a significance of 81 per cent using the McElhinny (1964) fold test. On the McFadden (1990) fold test, however, the directions *in situ* show a correlation with tectonic dip with the statistic $\xi = 6.685$, whereas after unfolding $\xi = 2.638$, and there is no apparent correlation. For $N = 22$ the 95 per cent significance point for ξ is 5.460 and at 99 per cent is 7.662. Therefore the fold test is positive at the 95 per cent level. No other combination of vectors listed in Table 2 for the Feixianguan Formation either passes or fails the McFadden (1990) fold test.

In these red sandstones when components are observed directly below and above 600 °C, we have interpreted them according to the classical experiments of Roy & Park (1974) on redbeds from North America. The first component (*A*) removed on thermal cleaning up to 600 °C is presumed to be a CRM from either magnetite or specular haematite. The high-temperature component (*B*) is presumed to be a CRM formed during redbed diagenesis. Because the normal and reverse *B* components are not statistically antiparallel it seems likely that there are two CRMs acquired at different times as also found by Roy & Park (1974). Unfortunately analyses of these results have only now been undertaken 10 years after the initial measurements were made and we are not in a position at this stage to carry out chemical demagnetization experiments to confirm our interpretation. If our interpretation is correct then it is not possible to use our results for magnetostratigraphy. Also it is likely that CRM due to haematite was formed in recent times providing a small overprint from the present field direction that cannot be distinguished from the ancient CRM. In this case the contamination would affect the reverse *B* component most. With this in mind our view is that the difference of 165.7° between the normal and reverse *B* components overestimates the departure from 180° and does not have any real significance with respect to the length of time over which diagenesis occurred. Clearly reversals of

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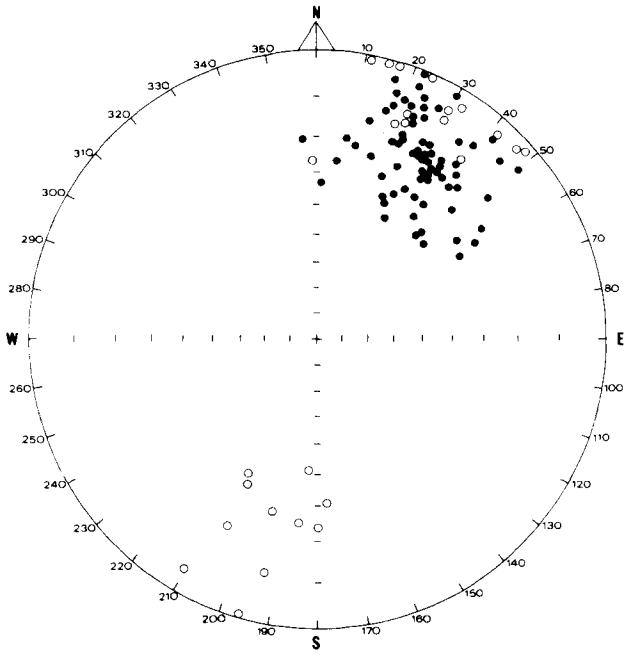


Figure 7. Equal-angle stereographic projection of sample directions from the Feixianguan Formation plotted in stratigraphic coordinates. Symbols are as defined in Fig. 5.

the magnetic field occurred during diagenesis and the predominance of a normal component below 600 °C suggests that early deposition occurred in a field of predominantly normal polarity. Later both normal and reverse components were acquired during diagenesis. Because the normal polarity *A* and *B* components are identical and the normal *B* component passes the fold test, we regard the best esimate of the palaeomagnetic field at the time of deposition and diagenesis of the Feixianguan Formation as being the overall mean direction of all the *A* and *B* components irrespective of sign. These directions are illustrated in Fig. 7. Again this was the procedure adopted by Roy & Park (1974).

The overall mean normal and reverse directions are 162.8° apart (Table 2) and therefore fail the reversals test of McFadden & McElhinny (1990). This is essentially a consequence of the fact that the reverse *B* component is not on average antiparallel to the normal *B* component and that all the components were acquired over several million years duing deposition and subsequent diagenetic processes. These processes probably occurred over several million years so that the time-averaged palaeomagnetic field for the entire formation will be well represented by the mean of all the normal and reverse vectors observed. The values listed is Table 2 show that inclusion of the 13 reverse vectors in the overall mean changes its direction by only 2.2°.

MAGNETOSTRATIGRAPHY

In the original report on the Emeishan Basalt from the Emei Mountain region (McElhinny *et al.* 1981) we assumed that the basalts lay within the Kiaman reversed superchron and thus their magnetization must be of reverse polarity. Zhao & Coe (1987, 1989) have pointed out that this interpretation is inconsistent with results from Triassic rocks and also with the fact that both normal and reverse polarities have been observed in the Emeishan Basalts from southern Sichuan and Yunnan Provinces. Stratigraphic considerations now place the basalts within the Longtanian Stage of the new 1989 Geologic Time Scale of Harland *et al.* (1990). This stage corresponds with the Tatarian Stage of the Russian Upper Permian sequences. Irving & Pullaiah (1976) have suggested, on the basis of Russian palaeomagnetic data, that the close of the Kiaman reversed superchron occurs at approximately the mid-Tatarian. This is precisely where the Longtanian Stage occurs on the Harland *et al.* (1990) time scale. Steiner *et al.* (1989) have recently summarized Permo–Triassic magnetostratigraphy worldwide comparing with their own sections sampled in the Sichuan Basin. There appear to be at least three normal polarity zones in the Late Permian within which the Emeishan Basalts could have been extruded. It seems clear now that the basalts are actually slightly younger in age than the Kiaman reversed superchron. It is now clear that the directions we observe are more logically taken to be of normal polarity, a decision that also results in a smooth polar wander path.

Our results from the Feixianguan Formation are dominated by normal polarity measurements. Although we do not believe these results can be used for detailed magnetostratigraphy, our interpretation of the data suggests that initial deposition occurred in a field of predominantly normal polarity. The results of both Heller *et al.* (1988), Steiner *et al.* (1989), Haag & Heller (1991) and Ogg & Steiner (1991) all show that the early part of the Griesbachian Stage at the base of the Lower Triassic is entirely of normal polarity. This is consistent with our sampling starting at the base of the Lower Triassic. A normal polarity chron is observed by Heller *et al.* (1988) and Steiner *et al.* (1989) in their measurements of the carbonate facies at the base of the Feixianguan Formation in the Sichuan Basin. Thus our data from the Feixianguan Formation red sandstones provide additional support for this Early Triassic magnetostratigraphy.

POLE POSITIONS

Chan, Wang & Wu (1984) have pointed out that the area sampled might have undergone a local rotation because it lies close to the Longmenshan thrust zone (Fig. 1 inset). Indeed McElhinny (1985) also suggested that this could be a possible interpretation of the data. Zhao & Coe (1987, 1989) and Enkin *et al.* (1992) have both reviewed the extensive studies that have been undertaken on the Emeishan Basalts throughout southwest China. Both authors have concluded that there are no differences between results from the various regions that imply any effects from local rotations in the Emei Mountain region due to the Longmenshan thrust zone. Enkin *et al.* (1992) show that local rotations in data from the Emeishan Basalts only occur in the vicinity of the Red River and Xianshuihe Faults. Other results including those from Emei Mountain are coherent with those from the Sichuan Basin. The proposals of Chan *et al.* (1984) and of McElhinny (1985) were both rather speculative and can now confidently be rejected.

Table 3 summarizes the most reliable Late Permian to Early Triassic pole positions that have been obtained from the South China Block. It should be noted that out results

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Table 3. Palaeomagnetic Poles for the Late Permian and Early Triassic from the South China Block.

Formation	No. Sites* or Samples*	Pole Position Lat N	Long E	dp	dm	Reference
Early Triassic						
Feixianguan	101*	56.2	227.6	1.7	3.4	Herein
Jialingjiang/ Feixianguan HPT	62*	38.5	209.8	1.7	3.3	(7)
Feixianguan HT	26*	38.7	211.8	3.2	6.3	(7)
Feixianguan	13*	43.3	211.7	3.3	6.4	(8)
Jialingjiang/ Feixianguan	4*(49*)	37.6	218.8	6.2	12.3	(8)
Jialingjiang	2*(16*)	56.8	212.6	3.2	5.9	(9)
Yelang	9*(47*)	46.3	219.2	5.7	11.2	(10)
Yinkeng	13*	43.6	222.2	9.6	9.6	(11)
MEAN POLE	*8*	*45.3*	*216.4*	*K=86.3*	*A_{95}=6.0°*	
Late Permian						
Emeishan Basalt	24*(125*)	54.5	254.3	1.8	3.5	Herein
Emeishan Basalt	32*(44*)	49.9	252.2	1.9	3.7	(1)
Emeishan Basalt	15*(15*)	53.2	233.9	6.3	12.6	(2)
Emeishan Basalt	9*(52*)	54.6	243.4	5.1	10.1	(3)
Xuanwei	12*	57.0	260.9	4.3	8.5	Herein
MEAN POLE	*5*	*54.2*	*248.8*	*K=149.3*	*A_{95}=6.3°*	
Permo-Triassic						
Changxing	16*	51.1	230.3	3.9	4.3	(4)
Changxing	13*(42*)	50.7	204.8	5.4	8.6	(5)
Feixianguan/ Dalong	114*	47.9	225.1	2.8	5.4	(6)
Feixianguan/ Changxing GPW	38*	47.2	226.3	2.1	4.2	(7)
Feixianguan/ Changxing LP	17*	45.8	225.2	3.5	7.0	(7)
MEAN POLE	*5*	*48.9*	*222.5*	*K=137.9*	*A_{95}=6.5°*	

dp, dm are the semi-axes of the ovals of 95 per cent confidence about the mean pole position. K is the Fisher (1953) precision parameter and A_{95} is the radius of the circle of 95 per cent confidence about each of the group mean poles.
(1) Zhao & Coe (1989); (2) Huang *et al.* (1986);
(3) Liu, Zhu & Jing (1987); (4) Li & Wang (1987); (5) Lin (1984);
(6) Heller *et al.* (1988); (7) Steiner *et al.* (1989);
(8) Enkin *et al.* (1992); (9) Zhu, Hao & Zhao (1988);
(10) Opdyke *et al.* (1986); (11) Dobson *et al.* (1992).

from the Emeishan Baslat, Xuanwei and Feixianguan Formations now supersede those previously published by McElhinny *et al.* (1981) and by Ma & Zhang (1989). Only those results with (1) a minimum of five flows (Emeishan Basalt) or 10 samples (sedimentary rocks) and (2) extensive demagnetization procedures have been included. We omit those results from the Emeishan Basalt that have been subjected to rotations near the Red River and Xianshuihe Faults (Enkin *et al.* 1992). The results have been divided into three groups as Late Permian (Longtanian to Changxingian), Permo–Triassic (Changxingian to Griesbachian) and Early Triassic (Griesbachian to Nammalian). Lin (1984) and Li & Wang (1987) assigned the Changxing Formation to the Early Triassic whereas on the 1989 Geologic Time Scale of Harland *et al.* (1990) the Changxiangian is now the latest stage of the Permian. Both Heller *et al.* (1988) and Steiner *et al.* (1989) studied sections that contain the Permo–Triassic boundary. In Table 3 we have grouped all of these studies together as a Permo–Triassic group and included all those studies that are overwhelmingly from Early Triassic sections into an Early Triassic group. The means for these groupings are shown in Fig. 8 with their circles of 95 per cent confidence.

The Late Permian group of poles is distinct from both the Permo–Triassic and Early Triassic groups. The Late Permian mean pole differs by 17.1° from the Permo–Triassic mean pole—a difference that is clearly significant (Fig. 8). The Permo–Triassic mean pole in turn differs by only 5.5° from the Early Triassic mean—a difference that is not significant. Thus there was a rapid shift with respect to South China during the latest Permian. The difference between the Late Permian and Early Triassic poles of the South China Block has been discussed fully by Enkin *et al.* (1992) who show that this is not a structural effect because of the predominance of Late Permian poles from the Emeishan Basalt outside the Sichuan Basin but a real effect from apparent polar wander. Note that our own results from the Emei Mountain region indicate a polar shift of 15.2° between the Emeishan Basalt and the overlying Feixianguan Formation similar to the overall results that includes data from other regions.

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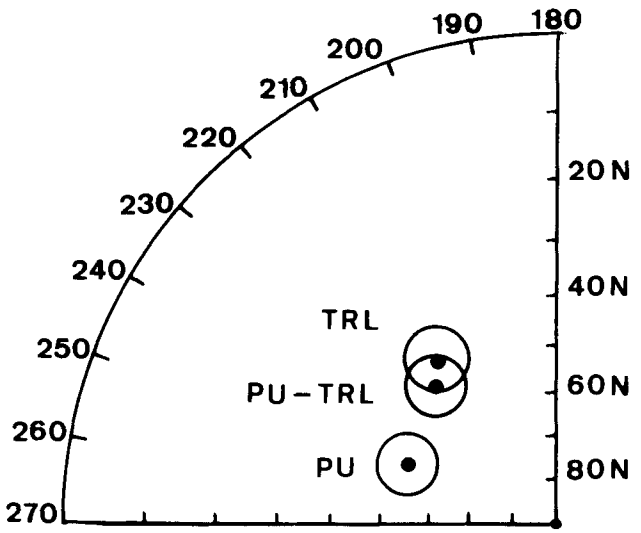


Figure 8. Mean Upper Permian to Lower Triassic pole positions for South China with circles of 95 per cent confidence. Equal-angle polar stereographic projection of the 180°–270° quadrant on the Earth's surface. Groupings are as given in Table 3 (PU: Upper Permian, PU-TRL: Permo–Triassic, TRL: Lower Triassic).

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Sichuan led by Chief Geologist Tao Xien Lin, geologists Tsao Zhu Yu and Xu Xiao Aun accompanied by Chuan Zhon Hai from the Chengdu Institute. They prepared detailed geological sections with accurate measurements of strike and dip of the strata at all points that were invaluable for making precise structural corrections. We gratefully acknowledge this assistance.

REFERENCES

Burrett, C. F., 1974. Plate tectonics and the fusion of Asia, *Earth planet. Sci. Lett.*, **21,** 181–189.

Chan, L. S., Wang, C. Y. & Wu, X. Y., 1984. Paleomagnetic results from some Permian–Triassic rocks from Southwestern China, *Geophys. Res. Lett.*, **11,** 1157–1160.

Dobson, J. P., Heller, F., Li, Z. X. & Mauritsch, H., 1992. Paleomagnetic and rock magnetic investigations of the Changxing Permo–Triassic, Changxing, China, *Geophys. Res. Lett.* in press.

Embleton, B. J. J., Zhang, Z., Ma, X., McElhinny, M. W. & Li, Z. X. Magnetostratigraphy of the Permo–Triassic boundary section near Taiyuan, Shanxi Province, China, in preparation.

Enkin, R. J., Courtillot, V., Leloup, P., Xing, L., Zhang, J. & Zhuang, Z., 1992. The paleomagnetic record of Permo–Triassic rocks from the South China Block, *Geophys, Res. Lett.*, **19,** 2147–2150.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, A **217,** 295–305.

Haag, M. & Heller, F., 1991. Late Permian to Early Triassic magnetostratigraphy, *Earth planet. Sci. Lett.*, **107,** 42–54.

Hamilton, W., 1970. The Uralides and the motion of the Russian and Siberian platforms, *Geol. Soc. Amer. Bull.*, **81,** 2553–2576.

Harland, W. B., Armstrong, R. L., Cox, A. V., Craig, L. E., Smith, A. G. & Smith, D. G., 1990. *A geologic time scale 1989,* Cambridge University Press, Cambridge.

Heller, F., Lowrie, W., Li, H. & Wang, J., 1988. Magnetostratigraphy of the Permo–Triassic boundary section at Shangsi (Guangyuan, Sichuan, China), *Earth planet. Sci. Lett.*, **88,** 348–356.

Huang, K., Opdyke, N. D., Kent, D. V., Xu, G. & Tang, R., 1986. Further paleomagnetic results from the Permian Emeishan Basalt in SW China, *Kexue Tongbao,* **31,** 1192–1201.

Irving, E. & Pullaiah, G., 1976. Reversals of the geomagnetic field, magnetostratigraphy, and the relative magnitude of paleosecular variation in the Phanerozoic, *Earth Sci. Rev.*, **12,** 35–64.

Kirschvink, J. L., 1980. The least square line and plane and the analysis of paleomagnetic data, *Geophys. J.R. astr. Soc.*, **62,** 699–718.

Li, H. & Wang, Z., 1987. Magnetostratigraphy of the Permo–Triassic boundary sections at Changxing, Zhejiang Province, China, *Abstr. Third Nat. Symp. Paleomagnetism,* Guangzhou, China.

Lin, J., 1984. The apparent polar wander paths for the North and South China blocks, *PhD thesis,* University California, Santa Barbara.

Liu, C. R., Zhu, R. & Jing, Z., 1987. Paleomagnetic study of Late Permian Emeishan Basalts from Panxi and its neighbouring areas, in *Contributions to Panxi Rift, China* 2, pp. 194–200, ed. Zhang, Y., Geological Publishing House, Beijing.

Ma, X. & Zhang, Z., 1989. The palaeomagnetic study of Permian rocks from Sichuan Emei region and Shanxi Taiyuan region (in Chinese), *Acta Geophysica Sinica,* **32,** (Suppl. 1), 451–465.

McElhinny, M. W., 1964. Statistical significance of the fold test in palaeomagnetism, *Geophys. J.R. astr. Soc.*, **8,** 338–340.

McElhinny, M. W., 1966. An improved method for demagnetizing rocks in alternating magnetic fields, *Geophys. J.R. astr. Soc.*, **10,** 369–374.

McElhinny, M. W., 1985. Permian paleomagnetism of the western Yangtze Block, China: a reinterpretation, *J. Geodynamics,* **2,** 115–117.

McElhinny, M. W., Luck, G. R. & Edwards, D., 1971. Large volume magnetic field free space for thermal demagnetization and other experiments in palaeomagnetism, *Pure appl. Geophys.*, **90,** 126–130.

McElhinny, M. W., Embleton, B. J. J., Ma, X. & Zhang, Z., 1981. Fragmentation of Asia in the Permian, *Nature,* **239,** 212–216.

McFadden, P. L., 1990. A new fold test for palaeomagnetic studies, *Geophys. J. Int.*, **103,** 163–169.

McFadden, P. L. & Jones, D. L., 1981. The fold test in palaeomagnetism, *Geophys. J.R. astr. Soc.*, **67,** 53–58.

McFadden, P. L. & McElhinny, M. W., 1990. Classification of the reversals test in palaeomagnetism, *Geophys. J. Int.*, **103,** 725–729.

McFadden, P. L., Ma, X., McElhinny, M. W. & Zhang, Z., 1988. Permo–Triassic magnetostratigraphy in China: northern Tarim, *Earth planet. Sci. Lett.*, **87,** 152–160.

Ogg, J. G. & Steiner, M. B., 1991. Early Triassic magnetic polarity time scale—integration of magnetostratigraphy, ammonite zonation and sequence stratigraphy from stratotype sections (Canadian Arctic Archipelago), *Earth planet. Sci. Lett.*, **107,** 69–89.

Opdyke, N. D., Huang, K., Xu, G., Zhang, W. Y. & Kent, D. V., 1986. Paleomagnetic results from the Triassic of the Yangtze platform, *J. geophys. Res.*, **91,** 9553–9568.

Roy, J. L. & Park, J. K., 1974. The magnetization process of certain red beds: vector analysis of chemical and thermal results, *Cand. J. Earth. Sci.*, **11,** 437–471.

Schmidt, P. W., 1982. Linearity spectrum analysis of multicomponent magnetizations and its application to some igneous rocks from south–eastern Australia, *Geophys. J.R. astr. Soc.*, **70,** 647–665.

Senanayake, W. E. & McElhinny, M. W., 1981. Hysteresis and susceptibility characteristics of magnetites and titanomagnetites: interpretation of results from basaltic rocks, *Phys. Earth planet. Inter.*, **26,** 47–55.

Steiner, M., Ogg, J., Zhang, Z. & Sun, S., 1989. The Late Permian/Early Triassic magnetic polarity time scale and plate motions of South China, *J. geophys. Res.*, **94,** 7343–7363.

Tapponnier, P., Peltzer, G. & Armijo, R., 1986. On the mechanics of the collision between India and Asia, in *Collision Tectonics,* pp. 115–157, eds Coward, M. P. & Ries, A. C., Geol. Soc. London Spec. Publ.

Yuan, H. & Li. Z., 1985. Preliminary study of isotopic geochronology of the magmatic rocks from the Panxi paleorift, in *Monograph on the Panxi rift zone, China,* pp. 241–251, ed. Zhang, Y. X., Geol. Publ. House, Beijing, China.

Zhao, X. & Coe, R. S., 1987. Palaeomagnetic constraints on the collision and rotation of North and South China, *Nature,* **327,** 141–144.

Zhao, X. & Coe, R. S., 1989. Tectonic implications of Permo–Triassic paleomagnetic results from North and South China, in *Deep structure and past kinematics of accreted terranes,* ed. Hillhouse, J. W., Am. geophys. Un. Geophys. Monogr., **50,** 267–283.

Zhang, Z. M., Liou, J. G. & Coleman, R., 1984. An outline of plate tectonics of China, *Geol. Soc. Am. Bull.*, **95,** 295–312.

Zhu, Z., Hao, T. & Zhao, H., 1988. Paleomagnetic study on the tectonic motion of Pan-Xi block and adjacent area during the YinShi–Yanshan period, *Acta Geophys. Sinica,* **31,** 420–431.

Zijderveld, J. D. A., 1967. A. C. Demagnetization of rocks: analysis of results, in *Methods in Palaeomagnetism,* pp. 254–286, eds Collinon, D. W., Creer, K. M. & Runcorn, S. K., Elsevier, Amsterdam.

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